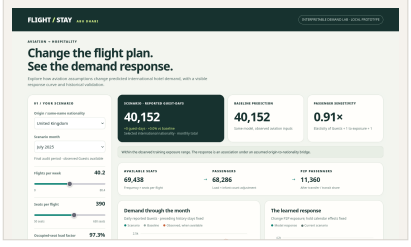
PROPOSAL / WORKING PROTOTYPE

Flight / Stay

A tool that links flights to hotel demand in Abu Dhabi.

EXECUTIVE SUMMARY

We built an offline simulator for the markets that already have flights. Change the flights, seats, load factor or connecting share. The simulator shows the change in hotel demand.
We held back June–July 2025. Error per nationality fell from **35.0%** to **27.7%**. The **35.0%** comes from a calendar model with no flight data. The slider response still needs a causal test.



The prototype runs now. Slide 8 shows the full view.

Prepared by [Team name] · [University / universities] · October 2026

Who we are and what we built

STUDENT TEAM

[Name 1] Data cleaning and checks

[Name 2] Models and validation

[Name 3] Interface and demo

Replace these names, these roles and the university before you submit.

CURRENT STATUS

- Our two notebooks run from start to end.
- We compared five readable models on a forward time split.
- The simulator is one HTML file. It runs with no server.
- The browser sliders and Python give the same numbers.

We want a first tool that a planner can question. We do not want a forecast that nobody can explain.

WHY THIS MATTERS

If flights change, what happens to hotel demand?

Airlines and hotels plan from different files. One seat can hold a transfer passenger, a returning resident or a hotel guest. So a direct seat-to-room number would be wrong.

QUESTIONS TO TEST

- What if the flights per week go up or down?
- What if an airline uses a larger aircraft?
- Which months and markets moved most in the past data?

The challenge asks for a tool that works, shows its steps and has a back-test. Source: ATP 2026 DCT Abu Dhabi challenge statement.

The two files count countries differently

WHAT THE FILES CONTAIN

- The flight file gives daily seats, passengers and connections.
- The hotel file gives daily *Guests* by nationality.
- The hotel file has 45 nationalities. 33 match a flight origin by name.
- The *Guests* numbers stop in July 2025.

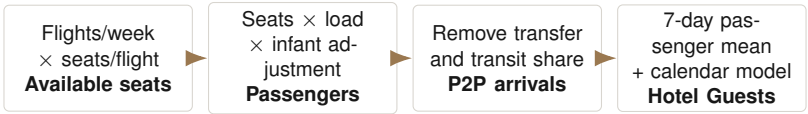
OUR STATED ASSUMPTIONS

- We link an origin to a nationality by name. This is an assumption, not a measurement.
- The other 12 nationalities get a calendar forecast. Their flight sliders stay off.
- We call the monthly sum of *Guests* reported guest-days. We cannot separate capture, length of stay and occupancy.

We drop the days with no hotel number. We start in 2023 because the 2022 flight rows are monthly, not daily.

How the simulator works

FIRST PLAIN ARITHMETIC, THEN A FITTED RESPONSE



INPUTS

Market, month, flights per week, seats per flight, load factor and transfer/transit share.

OUTPUTS

Predicted guest-days, the change from the real flights, the daily profile, the learned response and a warning for extreme inputs.

The fitted response holds the step from visitor to hotel guest. We cannot measure that step on its own.

We chose the simpler model on evidence

FIVE CANDIDATES

1. Calendar-only baseline
2. Proportional passenger chain
3. Monotonic regression **(selected)**
4. Monotonic GAM, 7-day exposure
5. Monotonic GAM, stay-kernel exposure

The two GAMs scored a little worse on the selection months. The extra curve gave no gain, so we left it out.

READABLE RESPONSE

$$\log(1 + \widehat{G}) = \alpha_m + \beta_m \log(1 + P_7) + \text{month} + \text{weekday} + \text{trend}$$
$$\beta_m \geq 0$$

Each matched market gets one coefficient. The coefficient cannot go below zero, so more passengers never lower demand. The seven-day mean lets an effect build up over days.

We chose the model on Jan–Apr 2025 only. If two models are within 1 point, we take the simpler one.

What the back-test shows

LOWER WMAPE IS BETTER

	Calendar	Flight model
Jan–Apr 2025 selection	31.9%	**25.0%**
Jun–Jul 2025 audit	35.0%	**27.7%**

WMAPE $= \sum |\text{prediction} - \text{observed}| / \sum \text{observed}$ over all nationality-days. We fit the models on Jan 2023–Dec 2024 after a 27-day warm-up. We keep May 2025 for the error bands.

OUR READING

7.2 points

Less error on the months we held back.

27.7%

Still too much error for one market on its own.

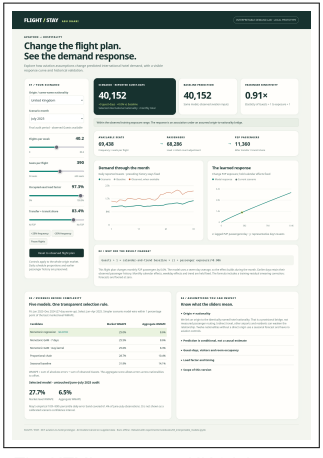
The daily total error is 6.5%, because one market's error can cancel another's. The test uses the flights that did operate. We changed no flights.

A working prototype, not just a diagram

WHAT A PLANNER CAN DO NOW

1. Pick a market and a month.
2. Move one slider: flights per week, seats per flight, load factor or connecting share.
3. Compare your scenario with the real flights. Read the passenger steps and the demand curve.
4. Watch for the warning when the scenario goes past the training range.

The prototype is one HTML file and needs no network. We check the browser math against Python.



The HTML prototype, UK / July 2025.

Where this could be useful

INTENDED IMPACT

- Show the flight and hotel assumptions in one place.
- Compare scenarios for current markets before a deeper study.
- Point to the months and markets that react most, for a person to review.

REAL-LIFE REQUIREMENTS

- Ask DCT what *Guests* counts. Ask for room counts, so we can show occupancy.
- Improve the link from flight origin to nationality with approved aggregate data.
- Reload the source data often. Track the error by market and by season.
- Keep new routes switched off until we can support them with evidence.

Today the slider shows an association under the assumptions above. It is not a proven effect of adding flights.

If shortlisted: test the hard part

DEMO AND TEST PLAN

1. Show three runs for one market and month: the baseline, 20% more flights, and a route cut.
2. Ask DCT to state the target definitions and the origin-to-nationality map.
3. Name the new test months and the real route changes first. Then check the error and the direction of the change.
4. Report the misses by market, by season and by how far the scenario goes past the data. Change the model only against months we have not used.

WHAT WE NEED TO IMPROVE

We need aggregate origin–nationality flows, a guest-night definition from DCT, event dates and room counts. With those we could test more of the seats-to-visitors-to-nights chain.

The prototype and the notebooks are ready for a live demo. Add the team details and the video link before you submit.

Sources: the ATP 2026 DCT Abu Dhabi challenge statement (challengeon.atrc.ae). The supplied DCT aviation and hotel files. Our domain analysis and model-comparison notebooks. We reproduced the results on 11 October 2026.